

REVIEWER REPORTS

Reviewer Comments:

Reviewer 1

Dear Reviewer, thank you for your comments and feedback on how to improve the manuscript. Each suggestion will be addressed below. All changes in the manuscript are highlighted in red.

Comments 1: I appreciate the authors' efforts to revise the manuscript and respond to my comments. The revised version has improved in several areas. However, after reviewing the manuscript and the supplementary materials, I believe that some important issues still need further clarification and correction.

In particular, the exclusion of electrical conductivity should be better justified because the SQIw is presented as an indicator of soil salinity. The exact sampling dates and the temporal matching between soil sampling and remote-sensing data also remain unclear. In addition, some performance metrics are calculated differently for the full and reduced models, and the same cross-validation folds appear to have been used for both model tuning and final evaluation. The permutation-importance analysis should also be checked because the sign may have been reversed.

There are also some inconsistencies that should be corrected, including CCC versus Pearson correlation, TRI versus TWI, and the limited description of the reduced model used to produce the final map. Finally, the objectives related to macronutrient supply and gypsum requirements should be supported with clearer quantitative results.

Section 1: In particular, the exclusion of electrical conductivity should be better justified because the SQIw is presented as an indicator of soil salinity.

We thank the reviewer for this observation. We consider it necessary to retain salinity as a central component of the study because the evaluated population predominantly corresponds to salt-affected soils, including saline–sodic soils according to the classification criteria used. However, ECe was not excluded a priori; rather, it did not show sufficient statistical capacity to discriminate the soil-quality gradient associated with the relative crop response represented by NDVI. This can be explained by the fact that salinity was widespread across the study area (the 25th percentile of ECe was 5.52 dS m⁻¹) whereas differences in soil quality were mainly associated with the sodicity gradient within these saline soils. Indeed, ECe showed an almost negligible correlation with SQIw ($r = -0.04$), whereas ESP ($r = -0.62$) and exchangeable Na ($r = -0.53$) showed much stronger and more consistent relationships with declining soil quality.

Therefore, the SQIw should not be interpreted as a direct indicator of salinity intensity, but rather as a soil-quality index developed for a predominantly saline and saline–sodic environment, where relative soil degradation was driven mainly by increasing sodicity. In addition, under some saline–sodic conditions, high electrolyte concentrations can partially attenuate the physical effects of Na by promoting flocculation and reducing clay dispersion; consequently, higher ECe does not necessarily correspond to poorer soil physical quality. This helps explain why the NDVI-related SQIw was not sensitive to ECe gradients, whereas it clearly discriminated against degradation associated with sodicity and alkalinity, with lower SQIw values observed in saline–sodic and alkaline saline–sodic soil groups.

Accordingly, we revised the Results, Discussion, and Conclusions sections to clarify the statistical basis for excluding ECe from the MDS and to explicitly distinguish the widespread saline condition of the study area from the sodicity gradient that more strongly determined the variation in SQIw.

Section 2: The exact sampling dates and the temporal matching between soil sampling and remote-sensing data also remain unclear.

We thank the reviewer for this observation. Soil sampling was conducted throughout the 2024–2025 agricultural season rather than during a single short sampling window, with most samples collected during 2024. This period was selected to represent the prevailing soil conditions of the dominant fruit-production systems in the study area. In these systems, particularly mango and banana, the second half of 2024 predominantly corresponded to vegetative development, whereas reproductive development and harvesting extended into the first half of 2025. Therefore, the Sentinel-2 period used to derive NDVI (January–December 2024) captured the main vegetative phase of the same 2024–2025 production cycle and was considered temporally representative of the crop condition associated with the soil survey.

Section 3: Some performance metrics are calculated differently for the full and reduced models.

Thank you for identifying this inconsistency. We revised the model evaluation workflow to ensure that all performance metrics are calculated using identical definitions and functions for the full and reduced models. Specifically, the calculation of relative error and the concordance correlation coefficient was standardized across all models. We also incorporated the reduced RF+RK model into the same five-fold Random CV framework used for OK, RF, and the full RF+RK model. All four models were subsequently compared using the Friedman test followed by the Nemenyi post-hoc test. Table 4 and the corresponding Results section have been updated accordingly

Section 4: The same cross-validation folds appear to have been used for both model tuning and final evaluation.

Thank you for highlighting this issue. We agree that using the same cross-validation folds for both hyperparameter tuning and final model evaluation could lead to optimistic performance estimates. To address this concern, the modelling workflow was revised using a nested cross-validation framework.

Specifically, a five-fold outer cross-validation was used exclusively for model evaluation. Within each outer training set, an independent five-fold inner cross-validation was performed for hyperparameter optimization using Optuna. Consequently, observations in each outer test fold were not used at any stage of hyperparameter selection. Final performance metrics were calculated exclusively from predictions on the held-out outer test folds and are reported as mean $\pm$ standard deviation across the five outer folds.

Section 5: The permutation-importance analysis should also be checked because the sign may have been reversed.

Thank you for pointing this out. We reviewed the permutation-importance calculation and confirmed that the sign had inadvertently been reversed. Because the scoring function was defined as negative RMSE, the permutation-importance output already represents the increase in RMSE after permutation ($\Delta\text{RMSE} = \text{RMSE}_{\text{permuted}} - \text{RMSE}_{\text{baseline}}$)

In addition, to ensure consistency with the revised model-evaluation framework, permutation importance was recalculated on the held-out outer test folds of the nested cross-validation, using the

RF model optimized exclusively within the corresponding outer training set. Predictor importance is now reported as the mean Δ RMSE across the five outer folds, with its standard deviation.

Section 6: There are also some inconsistencies that should be corrected, including CCC versus Pearson correlation.

Thank you for pointing out this inconsistency. We identified that Pearson's correlation coefficient had inadvertently been used for the reduced model, whereas Lin's concordance correlation coefficient (CCC) was used for the other models. We have corrected the analysis so that Lin's CCC is now calculated consistently for all models using the same function. The terminology has also been standardized throughout the manuscript, and "CC" has been replaced by "CCC (Lin's concordance correlation coefficient)."

Section 7: There are also some inconsistencies that should be corrected, including TRI versus TWI.

We thank the reviewer for identifying this inconsistency. Both TRI and TWI were initially included among the topographic covariates; however, TRI was the variable retained in the reduced model. The reference to TWI in the Discussion was a typographical inconsistency and has been corrected to TRI to ensure consistency with the model specification and Supplementary Figure 1.

Section 8: The limited description of the reduced model used to produce the final map.

Thank you for this observation. We have expanded the Methods section to provide a clearer description of the reduced RF+RK model and its use in generating the final SQIw map. Specifically, we clarified that the reduced predictor set was identified through permutation-importance analysis and that the reduced model was evaluated under the same nested cross-validation framework as the other predictive approaches.

Section 9: The objectives related to macronutrient supply and gypsum requirements should be supported with clearer quantitative results.

We thank the reviewer for this observation. We agree that the previous wording of the general objective overstated the scope of the quantitative analyses presented in the manuscript. The study did not directly estimate macronutrient supply rates or gypsum requirements in units such as kg ha^{-1} or t ha^{-1} . Rather, the SQIw and its associated soil properties were used to identify spatial gradients of soil quality and to delineate areas where differentiated nutrient and soil-amendment management may be required. Accordingly, we revised the general objective and the Conclusions to remove the implication that macronutrient supply and gypsum requirements were quantitatively estimated, thereby ensuring consistency between the stated objectives and the results presented.